

KITEWRIGHT RANGE · DESIGN DECISION RECORD

Kitewright Range round-3 decision, Kitewright family reconciliation

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	KWR-DDR-004	Version	0.1
Status	Draft	Date	2026-10-04
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-range		
Source	docs/decisions/0004-family-reconciliation.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-04	Amish Chadha	Amish's round-3 decision 10A carried out for Range

0004: Kitewright family reconciliation (round 3)

- **Date:** 2026-10-04
- **Status:** accepted. Decided by Amish Chadha on 2026-10-04: "For round 3, I agree with all your proposed recommendations" (round 3, item 10A).

■ Context

Range had assumed a 180 x 110 x 60 mm Core avionics box of 0.70 kg in its front bay, a separate 160 x 80 x 12 mm payload mount of 0.12 kg under the centre of gravity, a GNSS mast of 0.08 kg and XT90 leads on its packs. The Core, after its decision 33B, is a 240 x 150 mm plate with its own rail and pins that hangs under a frame's deck on a 220 x 130 mm M4 pattern, weighs 0.99 kg and takes AS150 plugs (KWR-CAL-001 v0.3 noted the gap and an estimate of 0.3 min less endurance from the mass alone).

■ Decision

10A: AS150 plugs everywhere; one mounting envelope for the Core shared by Lift and Range, with the Core's drawing as the reference; Range recomputed with the 0.99 kg Core; the Kitewright interface table in each repository's review note.

■ Carried out

Item	Was	Now
Kitewright Core	Avionics box in the front bay; separate payload mount plate under the CG	The Core hung under the floor at the CG to KWC-DWG-001 (<code>cad/src/core_envelope.py</code> , the same file as Lift's), its rail pointing aft so payloads slide in from the tail
Fuselage	140 mm wide; rear bulkhead at 585 mm	156 mm wide (150 mm inside, as the Core asks); floor cut with the 200 x 112 mm opening and four 4.3 mm holes; rear bulkhead at 645 mm
Core deck doubler (new)	None	6 mm birch, 300 x 150 mm, with the opening, four holes and M4 T-nuts, on the floor
Packs	Either side of the joiner on the floor	Front pack in the nose bay (60 to 198 mm), rear pack on the doubler behind the lid (506 to 644 mm); the centre of gravity stays at 28.3 % of the mean chord
GNSS and antennas	GNSS mast on the hatch	The Core's GNSS receiver on the hatch mast and its three antennas on the hatch, on extension cables
Payload envelope	150 x 90 x 100 mm under a mount plate	140 x 88 x 100 mm under a 184 x 128 x 5 mm payload shoe, inside the Core's 88 mm neck and clear of its pigtail plug
Pack leads	XT90	AS150

■ Results (KWR-CAL-001 v0.4)

- Take-off mass 11.35 kg (0.28 kg more); hover margin 48.3 % (R1 met on paper); endurance 30.6 min (R2 not met, accepted); range 41.5 km (R3 met on paper, 4 % margin); centre of gravity 28.3 % of the mean chord (R4).

- Value-engineering target: USD 5,000. Estimated cost of the constructable design: USD 4,314 (USD 686 under the target).
- The endurance falls 2.4 min, not the 0.3 min the Core's mass alone would cost: the doubler, wider fuselage and leads cost about 1.0 min and the drag of the Core hung under the floor about 1.2 min. LakeWatch's R5 is restated on 30.6 min.

■ Consequences

- No option in KWR-CAL-001 section H now meets R2 on paper; the moulded airframe with 6S4P packs comes closest (42.5 min).
- R3's margin is 4 %; a measured drag or thrust shortfall at TRL 4 could take it below 40 km.

Safety: The Core's pin knobs hang 140 mm above the ground and the payload 80 mm; land only on the four feet on clear ground. The Core power leads are soldered with every pack disconnected and checked for polarity before the first pack goes on.